Received: 28 May 2021 | Revised: 17 September 2021 | Accepted: 7 March 2022

DOI: 10.1111/jam.15565

ORIGINAL ARTICLE

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Perillaldehyde mitigates virulence factors and biofilm formation of *Pseudomonas aeruginosa* clinical isolates, by acting on the quorum sensing mechanism in vitro

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Abstract

Aim: The incidence of biofilm linked catheter-associated urinary tract infections is increasing worldwide and *Pseudomonas aeruginosa* is one of the major causes. Perillaldehyde (PLD): as a natural, widely used flavouring agent, has been reported to possess various pharmacological properties. We hypothesized that PLD can inhibit biofilm formation and virulence factor (VF) production by *P. aeruginosa* by hampering the quorum sensing (QS) system(s).

Methods and Results: Minimum inhibitory concentration (MIC) of PLD was assessed for standard strain and two multi-drug resistant catheter isolates of *P. aeruginosa* utilizing the microdilution method. Microtiter plate assay, crystal violet staining and scanning electron microscopy were used to evaluate the biofilm inhibition property. CFU was utilized to assess the antifouling property of PLD. Detection of VFs and expression analysis of virulence determinants were applied to investigate the anti-virulence activity. Gene expression and molecular docking studies were also executed to explore the QS inhibition and binding of PLD with QS receptors. In the present study, PLD has significantly inhibited biofilm formation and antivirulence activity at sub-MIC levels (2.5 and 3.5 mM) in all the tested strains. In addition, molecular docking studies revealed a significant affinity towards QS receptors.

Discussions: Perillaldehyde, being a non-toxic food flavouring agent, significantly inhibited biofilm formation and exhibited antifouling property. PLD exhibited significantly reduced levels of VFs ($p < 0.001$) and their respective genetic determinants ($p < 0.001$). Gene expression analysis and molecular docking studies confirmed the interactions of PLD to the QS receptors, indicating the plausible mechanism for the anti-virulence property.

Significance and Impact of Study: This study identified the anti-virulence potential of PLD and provided mechanistic insights. PLD can be a suitable, non-toxic candidate for countering biofilms and associated pathogens, contributing to the prevention of biofilm-associated nosocomial infections.

KEYWORDS

anti-microbial resistance, anti-virulence, catheter-associated urinary tract infections, *P. aeruginosa*, quorum sensing, virulence factors

INTRODUCTION

The use of urinary catheters in modern-day medicine contributes to the advancement of clinical outcomes in patients. However, these catheter surfaces are prone to microbial adhesion and biofilm development causing catheter-associated urinary tract infections (CAUTIs) (Redder et al., 2016). Biofilm-associated mineralization on these medical surfaces causes obstruction and loss of functionality by blocking the catheter lumen. Biofilm-associated infections are difficult to treat and eradicate, causing high mortality and morbidity (Hamilos, 2019).

With the ability to adhere effectively and efficiently to urinary catheter surfaces, opportunistic pathogens like *Pseudomonas aeruginosa* account for >39% of CAUTIs (Siddiq & Darouiche, 2012). Virulence factor (VF) production creates encouraging conditions for *P. aeruginosa* as they are masked by the host immune responses as well as antimicrobial treatment regimens (Moradali et al., 2017). The majority of these VFs including biofilm formation in *P. aeruginosa* is regulated by quorum sensing (QS) resulting in higher incidences of health threats like CAUTIs. Keeping this in view, there is an urgent need to attenuate virulence phenotypes of *P. aeruginosa*, rather than killing or inhibiting it (Chiang et al., 2018).

In pathogens, the production of VFs is governed by regulatory systems; in principle, the interference with these systems could alter their assembly. So, the microbial machinery included in the regulation of VFs production is an appropriate target for upsetting virulence (Fleitas Martínez et al., 2019). As an anti-virulence target, the QS system in *P. aeruginosa* is very well studied as *P. aeruginosa* utilizes QS schemes for regulating VFs. The QS machinery in *P. aeruginosa* is hierarchically arranged in which the *las* system triggers the other circuits; *rhl* and *pqs*.

The synthase protein, LasI and RhlI synthesize the autoinducers, 3-oxododecanoyl-L-homoserine lactone (3-oxo-C12-AHL) and *N*-butanoyl homoserine lactone (C4-HSL) which interacts with their associated transcriptional regulators, LasR and RhlR, respectively. Similarly, the Rhl system is revealed to regulate PQS (2-heptyl-3-hydroxy-4[1H]-quinolone) which is deliberated as a controlling connection between the *las* and *rhl* QS network. Under normal conditions, the QS circuitry in *P. aeruginosa* initiates the cell to cell communication mechanism via the 3-oxo-C12-AHL-LasR complex as the transcriptional regulator. This complex further regulates the other QS mechanisms and the expression of genetic determinants within the QS regulon (Kariminik et al., 2017).

Numerous compounds (natural and synthetic) have been comprehensively studied for their capability to inhibit VFs production by disrupting the QS system in *P. aeruginosa* (Viswanathan, Rathinam, & Suneeva, 2015).

Perillaldehyde (PLD, *p*-mentha-1, 8-dien-7-al) is a phytocompound abundantly found in the annual herb perilla (*Perilla frutescens*) and citrus fruit peels (Tian, Zeng, et al., 2015). In the modern era, PLD is widely used as a flavouring agent (Hobbs et al., 2016). Perillaldehyde is a monocyclic terpenoid containing functional groups like α,β-unsaturated aldehyde which gets metabolized through oxidation and excreted (WHO, 2017). In fact, PLD is "generally recognized as safe" (GRAS) by the Expert Panel of the U.S. Flavour and Extract Manufacturers Association (FEMA), was judged to be safe by the Food and Agriculture Organization of the United Nations (FAO)/World Health Organization (WHO, 2017) Joint Expert Committee on Food Additives (JECFA). Also, it has been designated unlikely to harm human health by the Japanese Ministry of Health, Labour and Welfare under the Food Sanitation Act (Hobbs et al., 2016; Honma et al., 2021). Perillaldehyde has been described to have numerous pharmacological properties such as significant hypolipidemic, anti-inflammatory, neuroprotective, antidepressant-like and anti-fungal effects (Ji et al., 2014; Omari-Siaw et al., 2016; Tian, Wang, et al., 2015; Tian, Zeng, et al., 2015). Perillaldehyde being a monoterpenoid compound containing an aldehyde functional group shared significant structural similarity with previously reported terpenoids (oleanolic aldehyde coumarate, ursolic acid, oleanolic acid, cassipourol and β-sitosterol) from various essential oils with potent anti-QS properties (Kiplimo et al., 2011; Rajkumari et al., 2019; Rasamiravaka et al., 2017). However, the effect of PLD on biofilm and other VFs production has not been studied. With similar structural features for an antagonistic action, it was hypothesized that PLD might bind to the QS receptors in a fashion comparable to their respective ligands.

Keeping in view of its inherent pharmacological effect, the biofilm inhibition and anti-virulence property of PLD were evaluated against *P. aeruginosa* PAO1 and 2 catheter isolates from patients with CAUTIs. The reverse transcription-quantitative PCR technique was used to analyse the effect of PLD on the expression of virulence determinants and QS-associated genes. Further, molecular docking studies were performed to analyse the plausible molecular target of PLD.

MATERIALS AND METHODS

Chemicals, bacterial strains and growth conditions

Perillaldehyde was purchased from Sigma-Aldrich and dissolved in dimethyl sulfoxide. *Pseudomonas aeruginosa* standard strain PAO1 and two clinical isolates of

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P. aeruginosa (RRLP1, GenBank ID: KR149278 and RRLP2, GenBank ID: KT309033) were used in this study (Rathinam et al., 2017). Fresh subcultures were prepared for each new experiment from the glycerol stocks maintained at −80°C and the bacterial cultures were subcultured till their mid-log phase (OD_{600} ~0.4). Further, for all experiments freshly prepared *P. aeruginosa* cells: ~1.8 $\times 10^8$ CFU/mL (OD_{600} ~0.08–0.1), from the mid-log phase cultures, were used.

Minimum inhibitory concentration determination

According to CLSI guidelines (2015), the micro-broth dilution method was employed to evaluate the effect of PLD on the growth of the test organisms (*P. aeruginosa* PAO1, RRLP1 and RRLP2). Overnight-grown test organisms; PAO1, RRLP1 and RRLP2 (adjusted to 0.5 McF) were grown in different concentrations of PLD ranging from 0.006 to 26 mM. The minimum inhibitory concentration (MIC) was determined by recording the absorbance (OD_{600}) following 24 h of incubation at 37°C.

Evaluation of growth curve

The planktonic cell growth assay as described previously (Rathinam & Viswanathan, 2018). Overnight-grown test organisms; PAO1, RRLP1 and RRLP2 (adjusted to 0.5 McF), were diluted 1:100 in LB with sub-MIC doses (2.5 and 3.5 mM) of PLD. Triplicate samples were grown in sterile 50 ml centrifuge tubes at 37°C with agitation (25.75 ***g***), and a 200 µl culture medium was carefully transferred to a microplate every 4 h for 24 h in total. Sterile medium served as control. The planktonic culture turbidity was read at 630 nm using a spectrophotometer (CHEMITO SPECTRASCAN UV2100; Thermo Scientific®).

Effects of PLD on the biofilm formation

The effect of PLD on the quantity of biofilm biomass formation was carried out using crystal violet (CV) staining (Costa et al., 2017; Mohamed et al., 2018; Rathinam & Viswanathan, 2018). Test organisms (PAO1, RRLP1 and RRLP2) were inoculated into trypticase soy broth (TSB) and cultivated with and without the sub-MIC doses of PLD (2.5 and 3.5 mM) at 37°C for 24 h, without agitation on microtitre plates with flat bottom wells. Biofilm formed was stained with CV (0.2%, w/v), washed with phosphate-buffered saline (PBS, pH 7.4), remaining CV was dissolved in 95% ethanol and spectrophotometrically analysed at OD_{570} (CHEMITO SPECTRASCAN UV2100; Thermo Scientific). The obtained data (OD_{570}) was corrected to the bacterial growth (OD_{600}). The percentage of biofilm inhibition was calculated using the formula:

$$\%\,\text{inhibition} = \left[(\text{Control OD} - \text{Test OD}) / \text{Control OD}\right] \times 100$$

Glass slides (1.0 cm × 1.0 cm), cut in a square shape, were placed in 24-well microtitre plates containing TSB with the test organisms (overnight-grown test organisms were diluted 1:100 in LB) and supplemented with and without sub-MIC dose of PLD (3.5 mM). Following incubation at 37°C for 24 h, the glass slides were rinsed twice with PBS (pH 7.4), stained using CV (0.2%, w/v) and viewed under a light microscope (MAGNUS; Magnus opto systems, equipped with SECUEASY digital high-resolution camera, India) under the magnification of 40×.

Biofilm development on urinary catheters in the presence of PLD

Urinary Foley catheters (Bard International, Inc.) were cut into 1.0 cm × 1.0 cm pieces and placed in the TSB medium with (3.5 mM) and without the supplementation of PLD. These were inoculated with 100 µl of overnight cultures of test organisms (10^5 CFU/ml of PAO1, RRLP1 and RRLP2) and incubated at 37°C. After every 24 h, catheter pieces were removed from the flask, rinsed thrice with sterile saline (pH 7.2) and transferred into a new flask containing fresh medium for 3 days. The biofilm formation on catheter pieces was studied using scanning electron microscopy (SEM). Catheter segments for SEM were preserved in cold acetone solution after rinsing with normal saline. Later, these segments were fixed in 2.5% glutaraldehyde and dehydrated using a series of ethanol wash (10%, 30%, 50%, 70%, 90% and 100%, v/v) before visualizing under a scanning electron microscope (ZEISS-EVO18; Carl Zeiss) (Abinaya & Gayathri, 2019).

Effect of sub-MIC doses of PLD on pre-formed biofilms

The method used by Saini et al. (2016), was employed with slight modifications to assess the effect of sub-MIC levels (2.5 and 3.5 mM) of PLD on preformed biofilms (Saini et al., 2016). Glass slides (1.0 cm × 1.0 cm), cut in a square shape, were allowed to form a conditioning film by exposing them to sterile TSB for a period of 4 h at 37°C. Later, these slides were inoculated separately with 10^5 CFU/ml of test organisms: PAO1, RRLP1 and RRLP2 per millilitre of TSB (with and without PLD) and incubated at 37°C for 24 h to allow biofilm formation. After incubation, glass slides (both PLD treated and untreated) were first rinsed with sterile saline (0.9% NaCl), and the adherent bacteria were detached by sonication; for 15–20 min at 50 Hz. After sonication, each glass slide was vortexed and the achieved bacterial suspension was serially diluted and spread plated on MacConkey Agar (HiMedia Laboratories Pvt. Ltd.). The plates were incubated at 37°C for 24 h and the bacterial count was enumerated using a colony counter.

Effects of PLD on the production of VFs

Production of pyocyanin, protease, elastase, pyoverdine, rhamnolipids, cell surface hydrophobicity (CSH), extracellular polysaccharides (EPS) and swarming motility by the tested organisms (PAO1, RRLP1 and RRLP2) in the presence and absence of sub-MIC doses of PLD (2.5 and 3.5 mM) was assessed using standard methods as described below. For all experiments freshly prepared *P. aeruginosa* cells: ~1.8×10^8 CFU/ml (OD_{600} ~0.08–0.1), from the mid-log phase cultures, were used. All the test organisms were grown in an LB medium with or without a sub-MIC dose of PLD (2.5 and 3.5 mM). The culture supernatants obtained were filtered and employed for the pyocyanin estimation. Three millilitres of supernatant was mixed with chloroform (5 ml) and the organic phase was collected and acidified using HCl (1 ml, 0.2 M). The absorbance (OD_{520}) was recorded to estimate the pyocyanin concentration (Rathinam et al., 2017). Cell-free culture supernatants (100 µl) of test organisms (treated with and without PLD) were added to 900 µl of boiled *Staphylococcus aureus* cells, and the absorbance (OD_{600}) was measured after an incubation of 60 min, to estimate LasA protease production (Kong et al., 2005). The obtained results were compared with that of the PLD-untreated culture supernatants. Cell-free culture supernatants (100 µl) of test organisms (treated with and without PLD) were supplemented with 900 µl of Elastin Congo Red (ECR) buffer containing 20 mg of ECR, and the LasB elastase activity was measured at OD_{495} (Ohman et al., 1980). Cell-free culture supernatant of the tested organisms (grown with and without PLD) was diluted 10-fold in Tris–HCl buffer (pH 7.4) and the relative concentration of pyoverdine was estimated based on the fluorescence analysis at an excitation of 405 nm and an emission of 465 nm (Cox & Adams, 1985). The 100 µl culture supernatants (treated and untreated with sub-MIC doses of PLD) were extracted using ethyl acetate and treated with orcinol reagent (900 µl, 0.19%). Further, the rhamnolipids was estimated at OD_{421} as per the method by Zhu et al. (2002). The obtained data was corrected to the OD_{600} to normalize for differences in the growth of tested strains. The bacterial cultures (PAO1, RRLP1 and RRLP2) that were treated and untreated with sub-MIC doses of PLD were mixed vigorously with toluene (1 ml) for 2 min. The absorbance of the aqueous phase was recorded at OD_{600} to determine the CSH (Chatterjee et al., 2017). The obtained data was corrected to the OD_{600} to normalize for differences in the growth of tested strains. The EPS produced by the sub-MIC doses of PLD-treated and -untreated PAO1, RRLP1 and RRLP2 biofilms were extracted using chilled ethanol (99%) and quantitatively estimated at OD_{490} using the phenol-sulphuric acid method (Nielsen, 2010). The obtained data was corrected to the OD_{600} to normalize for differences in the growth of tested strains. Swarm agar plates constituting 0.5% peptone, 0.2% yeast extract, 1% glucose and 0.5% bacto agar amended with and without PLD were prepared. An overnight culture of the test organisms (2 µl) was spot inoculated in the middle of the semi-solid agar plates and the migration distance was measured after 48 h of incubation (Chatterjee et al., 2017). The percentage inhibition was calculated using the formula:

$$\%\,\text{inhibition} = \left[(\text{Control OD} - \text{Test OD}) / \text{Control OD}\right] \times 100$$

RNA extraction and quantitative real-time PCR

The tested organisms (PAO1, RRLP1 and RRLP2) were grown in a sterile LB medium supplemented with (3.5 mM) or without PLD at 37°C for 24 h. Total RNA was extracted using TRIzol reagent (Sigma-Aldrich) per the manufacturer's protocol. The concentration and purity of extracted total RNA were determined by ultraviolet absorption (260/280 nm). The first-strand cDNA was generated from a purified mRNA sample using a Verso cDNA synthesis kit (Thermo Scientific) per the manufacturer's instructions. Quantitative real-time PCR (RT-qPCR) was carried out using the PrimeScript™ RT master mix (TaKaRa Bio, Inc.) with a Real-Time PCR instrument (StepOne™; Thermo Fisher Scientific). The reaction procedure was performed as follows: 95°C for 30 s, 40 cycles of 95°C for 5 s, 60°C for 5 s and a final melting curve analysis from 65 to 95°C (with increments of 0.5°C every 5 s) (Rathinam et al., 2017). All the RT-qPCR analyses were performed in quadruplicate.

Primer sequences for the VFs genes; *lasA* (encoding protease), *lasB* (encoding elastase), *phzA1* (encoding pyocyanin), *pvdA* (encoding pyoverdine), *rhlA* (encoding rhamnosyl transferase), *pslB* (encoding psl biofilm polysaccharide) and QS genes; *lasR rhlR* and *mvfR* (codes for QS receptors), *lasI* and *rhlI* (QS signal synthase genes) for PAO1, RRLP1 and RRLP2 for the test organisms were described in Table S1. The ribosomal gene 16s rRNA was chosen as a housekeeping gene to normalize the RT-qPCR data. Amplification profiles were analysed using StepOneDataAssist® software (Thermo Fisher Scientific), and cycle threshold (Ct) values for each target gene were normalized to the geometric mean of the Ct of 16s rRNA amplified from the corresponding sample. The fold change (RQ) of the target genes for each group to the control group was calculated using the $2^{-\Delta\Delta ct}$ method and compared to the control (without PLD) (Rathinam et al., 2017).

Molecular docking

The X-ray crystal structure of PqsR ligand binding complex was retrieved from Protein Data Bank (PDB ID: 4JVI) (Soukarieh et al., 2018). The target was prepared by using the Protein Preparation Wizard Schrödinger Suite 2018-4 for refining and optimizing the crystal structures, correcting the protonation states, adding missing hydrogen atoms and reconstructing incomplete side-chains. Ligand has prepared by using the Ligprep tool. The obtained files were thus used for docking simulations by employing Grid-based Ligand Docking with Energetics (GLIDE) (Release, 2018; Schrödinger, 2018).

To get further insight into the docking study, the protein-ligand complexes obtained from XP docking were further refined and carried out a post-docking calculation using the MM-GBSA module (Molecular Mechanics/Generalized Born Surface Area) available in Schrodinger. MM-GBSA is a constructive method for calculating the free energy of binding ($\Delta G_{[\mathrm{bind}]}$) which has a direct relationship with binding affinities. The calculation was performed by considering the solvation energies of the interacting molecules in conjunction with molecular mechanics (MM) based energies. The implicit solvent model (GB) was used to calculate the contribution of polar solvation energy and solvent accessible surface area (SA) was used to calculate the nonpolar part of the solvation energy. The free energy of binding was estimated using the following equation.

$$\Delta G_{(\mathrm{bind})} = E_{\mathrm{complex}}\,(\mathrm{minimized}) - \left(E_{\mathrm{protein}}\,(\mathrm{unbound, minimized})\right) + E_{\mathrm{ligand}}(\mathrm{unbound, minimized})$$

where $\Delta G_{(\mathrm{bind})}$ is the calculated binding free energy, E_{complex}(minimized) is the MM-GBSA energy of the minimized complex, E_{protein} (unbound, minimized) is the MM-GBSA energy of the minimized protein after separating it from its bound ligand and E_{ligand} (unbound, minimized)is the MM-GBSA energy of the ligand after separating it from the crystal complex and allowing it to relax.

Data analysis

All experiments were performed independently, and all the data except gene expression studies were represented as mean ± standard deviation (SD) with the p value of 0.05 being significant using GraphPad Prism 5. Results of the gene expression studies were represented as fold change ± SD with the p value of 0.05 being significant using GraphPad Prism 5. All the results of the VFs estimation assays, biofilm inhibition, effect on preformed biofilms and competition assays were analysed by two-way analysis of variance (ANOVA) with Tukey's multiple comparison post-tests. Comparisons were performed within the VF production in different PLD concentrations as 2.5 mM vs 3.5 mM. Gene expression results were analysed by two-way ANOVA with Sidak's multiple comparison post-tests in which comparisons were performed as PLD untreated versus PLDtreated (3.5 mM).

RESULTS

Determination of MIC and the effect of cell growth

The MIC, and minimum bactericidal concentration of PLD for the test pathogens (PAO1, RRLP1 and RRLP2) were determined as (6.8 mM) and (13.6 mM), respectively. As depicted in the growth curve (Figure S1), the sub-MIC level of PLD (5.5 mM) did not show any significant inhibitory effect on the growth profile of PAO1, RRLP1 and RRLP2 with respect to control. Hence, the sub-MIC levels of PLD: (2.5 mM) and (3.5 mM) were employed for all further experiments.

PLD exhibited significant biofilm inhibition against the tested organisms

In the present study, PLD-untreated biofilms of all the tested organisms exhibited greater biofilm production. Whereas sub-MIC levels of PLD (2.5 and 3.5 mM) significantly inhibited biofilm formation in PAO1, RRLP1 and RRLP2 compared to the respective controls. In addition, among the tested doses, 3.5 mM of PLD exhibited significant biofilm inhibition ($p < 0.001$) than 2.5 mM (Figure 1). The % inhibition of PLD (3.5 mM) on biofilm production by PAO1, RRLP1 and RRLP2 was found to be 65.6%, 67.9% and 64.4%, respectively. Further, the microscopic images of the CV stained biofilms (without PLD) formed on the glass slides revealed a dense, well-developed biofilm. However, 3.5 mM of PLD-treated slides exhibited significantly reduced biofilm formation with visibly hampered architecture. In addition, the presence of PLD hampered the structural organization with a loosely arranged biofilm matrix (Figure 2).

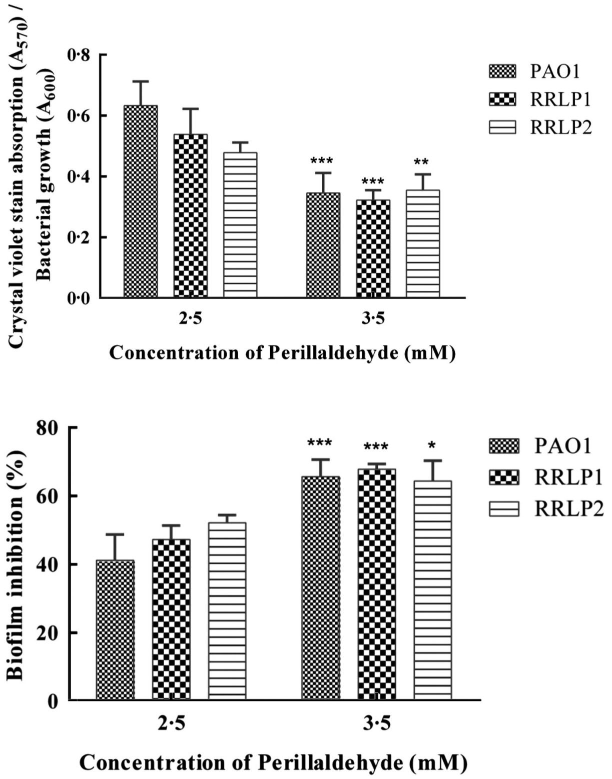

FIGURE 1 Effect of sub-MIC levels of perillaldehyde on biofilm formation by *Pseudomonas aeruginosa* PAO1, RRLP1 and RRLP2. The results are represented as mean ± SD ($n = 6$) and analysed by two-way ANOVA with Tukey's multiple comparison post test. Comparisons were performed within the biofilm production in different perillaldehyde concentrations as 2.5 mM versus 3.5 mM (*$p < 0.05$, ***$p < 0.001$). ANOVA, analysis of variance; MIC, minimum inhibitory concentration; SD, standard deviation

Because of the promising biofilm inhibition property, the effect of PLD on preformed biofilms was examined and enumerated using a colony counter (Figure 3). Glass slides were allowed to form initial conditioning films and further, the tested organisms were allowed to form biofilms in the presence and absence of sub-MIC dose of PLD. Biofilm-associated bacteria that were extracted from these segments were enumerated using serial dilution and spread plate technique. All the tested organisms exhibited elevated levels of viable biofilm-associated bacteria on control (PLD untreated) slides during the study period, enumerating an average of 8.17 Log CFU (PAO1), 9.2 Log CFU (RRLP1) and 9.5 Log CFU (RRLP2), respectively. On the contrary, reduced biofilm linked organisms (PAO1, RRLP1 and RRLP2) specifying reduced bacterial adhesion were observed with PLD-treated biofilms. Among the tested organisms the biofilm associated with PAO1 exhibited an enumeration of 5 Log CFU and 1.4 Log CFU for 2.5 and 3.5 mM ($p < 0.001$) of PLD, respectively. Among the clinical isolates used, RRLP1 and RRLP2 exhibited a significantly reduced biofilm bacterial load of 0.6 Log CFU and 0.7 Log CFU, respectively with 3.5 mM ($p < 0.001$) of PLD than 2.5 mM. These findings suggested that a sub-MIC dose of PLD could significantly eradicate biofilm-associated bacteria even after the conditional film formation.

Biofilm inhibition property was further appraised by SEM, where the biofilm formation by PAO1, RRLP1 and RRLP2 on the catheter segments (treated and untreated with 3.5 mM of PLD) was visualized (Figure 4). Control segments of all the tested organisms showed tightly packed, dense and matured biofilms with complex architecture. Notable extracellular matrix formation with thread-like structures was also present in these biofilms. On the contrary, PLD-treated segments exhibited architecturally

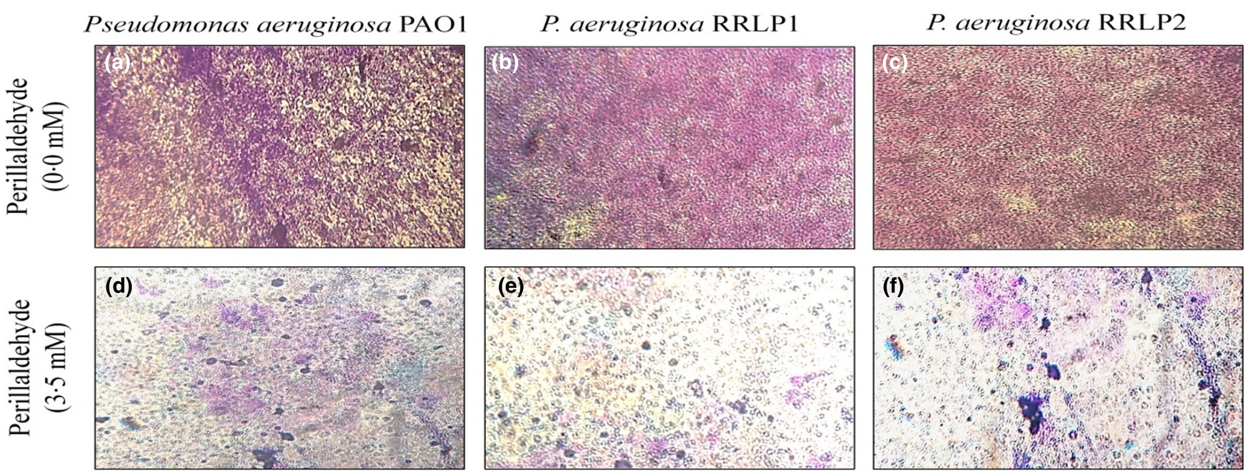

FIGURE 2 Light microscopic images (40×) of crystal violet stained biofilms at (0.0 mM) perillaldehyde (a) *Pseudomonas aeruginosa* PAO1; (b) *P. aeruginosa* RRLP1; (c) *P. aeruginosa* RRLP2 and at 3.5 mM of perillaldehyde; (d) *P. aeruginosa* PAO1; (e) *P. aeruginosa* RRLP1; and (f) *P. aeruginosa* RRLP2

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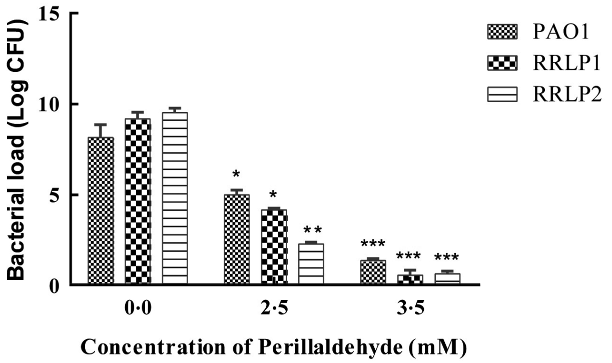

FIGURE 3 Antifouling property of sub-MIC doses of perillaldehyde (2.5 and 3.5 mM) against *Pseudomonas aeruginosa* PAO1, RRLP1 and RRLP2 (24 h). The results are represented as mean ± SD ($n = 3$) and analysed by two-way ANOVA with Sidak's multiple comparison post-test. The comparisons were performed control versus tested perillaldehyde concentrations (*$p < 0.05$, **$p < 0.01$, ***$p < 0.001$). ANOVA, analysis of variance; MIC, minimum inhibitory concentration; SD, standard deviation

simpler biofilms with reduced density of adherent bacterial cells, and complete inhibition of biofilm components.

Effect of sub-MIC level of PLD on the production of VFs in the tested organisms

With the sub-MIC levels of PLD (2.5 and 3.5 mM), significant suppression in the production of VFs by the tested organisms (PAO1, RRLP1 and RRLP2), was observed, as represented in (Figure 5a–c), respectively. With 3.5 mM of PLD, the percentage inhibition of pyocyanin production in the test organisms followed the order of PAO1 (67.86%, $p < 0.001$) > RRLP1 (61.54%, $p < 0.001$) > RRLP2 (37.52%, $p < 0.05$). Similarly, the % reduction in pyoverdine production was 62.86, 69.58 and 68.73 by PAO1, RRLP1 and RRLP2, respectively ($p < 0.001$).

Increased levels of extracellular enzymes like proteases and elastases directly affect the pathogenicity by inducing host tissue degradation, thereby facilitating bacterial invasion and proliferation. In the present study, PAO1 (Figure 5a) and the clinical isolates (RRLP1 and RRLP2) exhibited elevated protease and elastase (Figure 5b,c) enzyme levels in the absence of PLD (control), whereas 3.5 mM of PLD significantly reduced the production of these exo-enzymes ($p < 0.001$) than 2.5 mM. The % reduction in the LasA protease production was found to be higher with 3.5 mM of PLD as 67.34 (PAO1), 59.15 (RRLP1) and 50.00 (RRLP2). Similarly, an order of RRLP2 (65.18%) > RRLP1 (48.32%) > PAO1 (42.18%) was followed for the reduction in LasB elastase production with 3.5 mM of PLD.

Virulence factors like EPS, rhamnolipids, CSH, swarming motility and rhamnolipids are important factors for the initial stages of biofilm establishment aiding in biofilm matrix formation and microbial adhesion (Figure 5). Sub-MIC doses of PLD (2.5 and 3.5 mM) significantly inhibited all the biofilm-associated VFs in PAO1 (Figure 5a), RRLP1 (Figure 5b) and RRLP2 (Figure 5c). In fact among the tested doses of PLD, 3.5 mM exhibited more significant anti-virulence activity than 2.5 mM. Extracellular polysaccharides are highly hydrated biopolymers of microbial origin in which biofilm microorganisms are embedded. In addition, EPS form a matrix, which keeps the biofilm cells together and retains water. The effect of PLD (2.5 and 3.5 mM) on EPS production was evaluated and the obtained results revealed that 3.5 mM of PLD exhibited a significant % reduction ($p < 0.001$) of 66.12%, 64.41% and 65.54% in the EPS production by PAO1, RRLP1 and RRLP2, respectively in comparison with the 2.5 mM dose. The amphipathic glycolipid: rhamnolipids plays a major role in biofilm production by facilitating bacterial motility, increasing CSH and in maintaining open

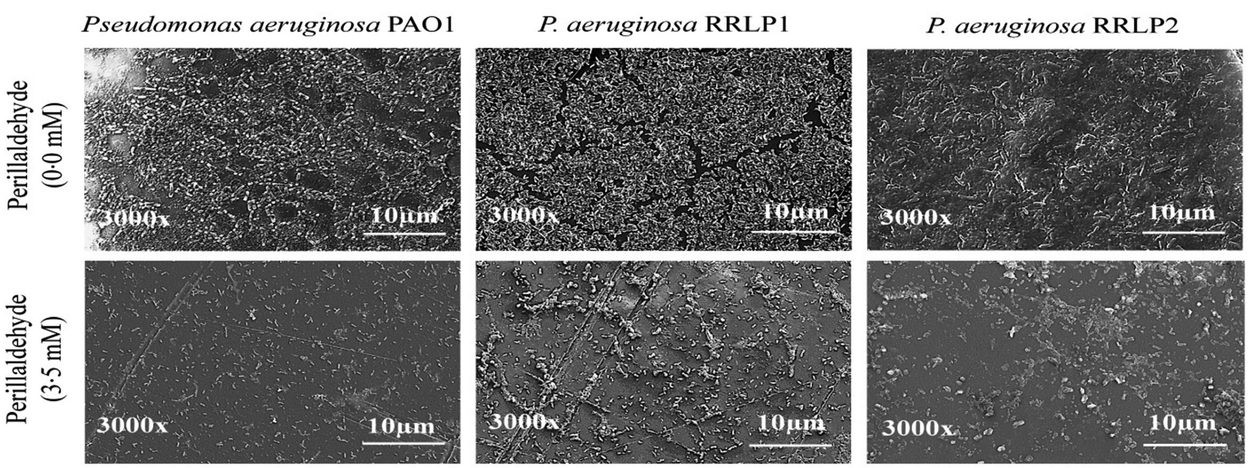

FIGURE 4 Scanning microscopic images of perillaldehyde-treated (3.5 mM) biofilms by *Pseudomonas aeruginosa* PAO1, RRLP1 and RRLP2

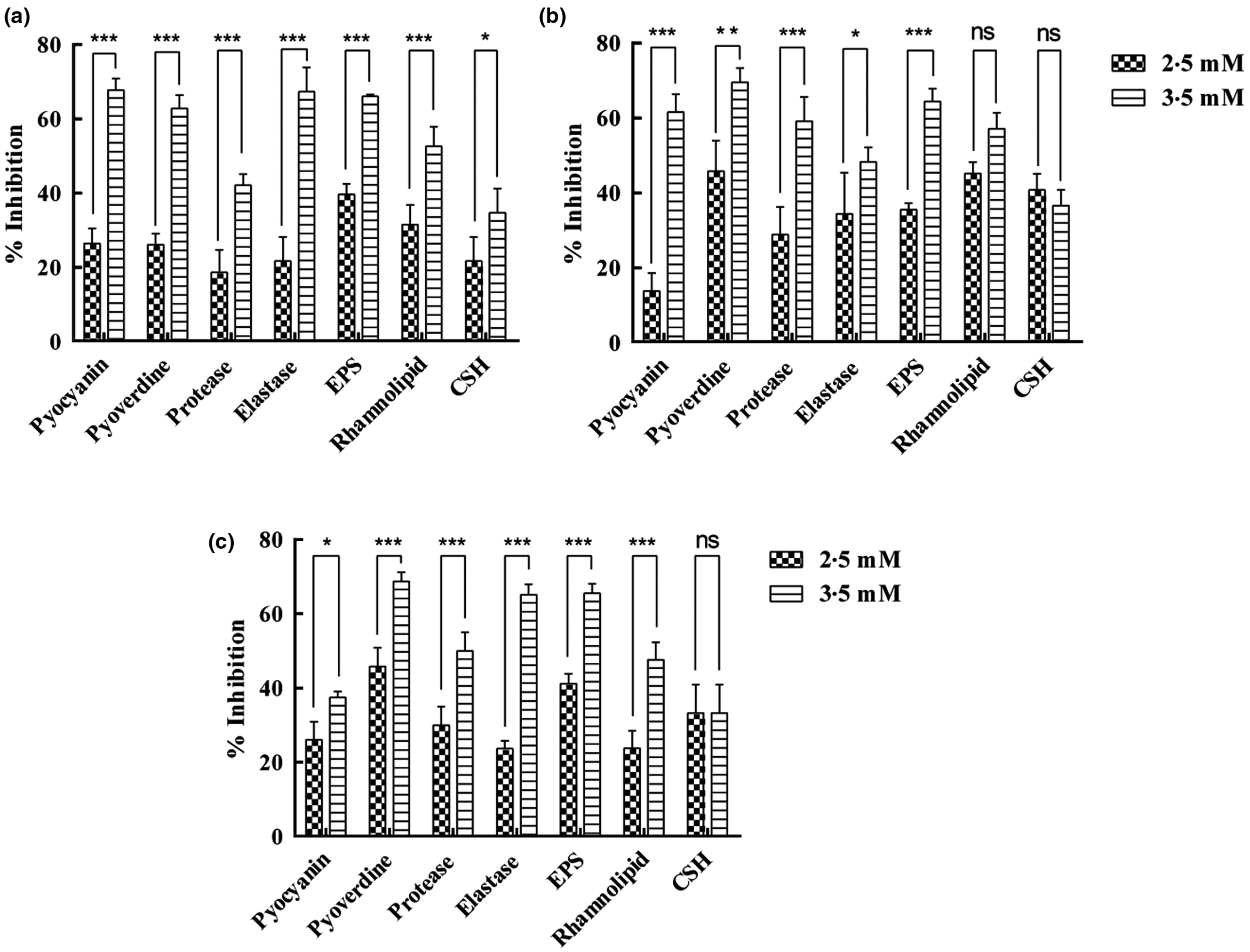

FIGURE 5 Effect of sub-MIC levels of perillaldehyde (2.5 and 3.5 mM) on virulence factor production. (a) PAO1; (b) RRLP1; and (c) RRLP2. The results were represented as mean ± SD ($n = 3$) and analysed by two-way ANOVA with Tukey's multiple comparison post-test. Comparisons were performed within the virulence factor production in different perillaldehyde concentrations (***$p < 0.001$, **$p < 0.01$, *$p < 0.05$, ns, not significant). ANOVA, analysis of variance; MIC, minimum inhibitory concentration; SD, standard deviation

channels thereby increasing the formation of micro as well as macro-colonies. Perillaldehyde-treated culture supernatants of tested organisms exhibited significantly reduced levels of rhamnolipids than PLD untreated (control). Among the measured levels of rhamnolipids production, maximum % reduction was observed as 52.6 (PAO1, $p < 0.001$), 57.14 (RRLP1, $p < 0.05$), and 47.62 (RRLP2, $p < 0.001$), respectively.

During the biofilm mode of life, the surface properties of the biofilm linked microbes become hydrophobic as the CSH is essential for bacterial adhesion. Furthermore, the swarming motility also aids in initial adherence and colonization of the substratum. This study evidences that a sub-MIC dose of PLD significantly decreased the CSH and swarming motility in all the tested organisms. As evidenced in the present study among the tested doses of PLD 3.5 mM dose exhibited a maximum % reduction in swarming motility (Table 1), whereas, both the tested PLD levels exhibited no significant difference in the CSH inhibition. The % reduction of uCSH for PAO1, RRLP1 and RRLP2 were 62.39, 58.45 and 50.00 respectively. In fact, the swarming motility inhibition by the 3.5 mM of PLD followed an order of 75.5% (RRLP2) > 73.8% (PAO1) > 73.75% (RRLP1) ($p < 0.001$).

PLD reduced the mRNA levels of virulence determinants and QS-associated genes

The effect of PLD (3.5 mM) on the key VFs determinants (*lasA*, *lasB*, *phzA1*, *pvdA*, *rhlA* and *pslB*) was monitored with RT-qPCR (Figure 6). The mRNA levels of all the tested VFs genes for PAO1, RRLP1 and RRLP2 were significantly ($p < 0.001$) decreased after sub-MIC dose of PLD (3.5 mM) in comparison with the control (PLD untreated), as represented in (Figure 6a–c). In PAO1, the psl biofilm polysaccharide gene *pslB* was observed a down-regulation with a relative fold level of 0.07 ± 0.03) ($p < 0.001$). The mRNA levels of genes coding for elastase (*lasB*) and enzyme for pyoverdine siderophore synthesis (*pvdA*), was also significantly reduced as 0.28 ± 0.04 and 0.20 ± 0.01 ($p < 0.001$) in

TABLE 1 Percentage inhibition of swarming motility by perillaldehyde (3.5 mM)

S. No.	Perillaldehyde concentration (mM)	Swarming motility diameter (mm)	Inhibition of motility (%)
1	*Pseudomonas aeruginosa* PAO1 (0.0 mM)	42.66 ± 2.5	0.0
2	*P. aeruginosa* PAO1 (3.5 mM)	11.17 ± 0.15	75.8***
3	*P. aeruginosa* RRLP1 (0.0 mM)	40.00 ± 2.0	0.0
4	*P. aeruginosa* RRLP1 (3.5 mM)	10.5 ± 0.5	73.75***
5	*P. aeruginosa* RRLP2 (0.0 mM)	44.0 ± 1.4	0.0
6	*P. aeruginosa* RRLP2 (3.5 mM)	10.77 ± 0.25	75.5***

Note: The results were represented as mean ± SD ($n = 6$) and analysed by two-way ANOVA with Tukey's multiple comparison post-test. Comparisons were performed within the virulence factor production in different perillaldehyde concentrations (***$p < 0.001$).

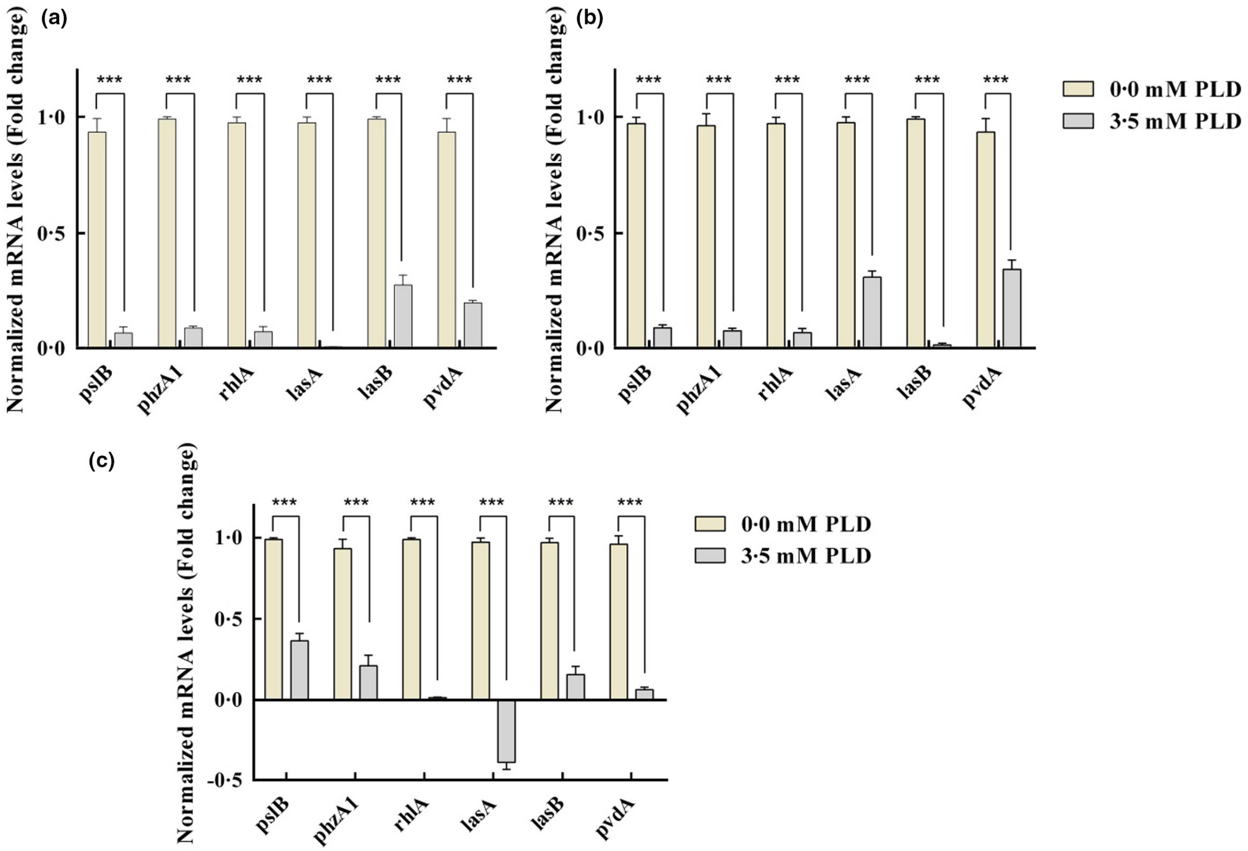

FIGURE 6 mRNA expression analysis of virulence genes after perillaldehyde (3.5 mM) treatment. (a) PAO1, (b) RRLP1 and (c) RRLP2. The results were represented as mean ± SD ($n = 4$) and analysed by two-way ANOVA with Sidak's multiple comparison post-test. Comparisons were performed as respective genes of perillaldehyde untreated (Control) versus perillaldehyde treated (***$p < 0.001$). ANOVA, analysis of variance; SD, standard deviation

PLD-treated PAO1 cells. Similarly, both the *phzA* and *rhlA* genes exhibited significantly reduced relative fold levels of 0.09 ± 0.01, and 0.076 ± 0.02, respectively (Figure 6a). Among the clinical isolates, PLD (3.5 mM) significantly reduced the fold expression levels of *lasA* (0.31 ± 0.03), *lasB* (0.019 ± 0.01), *phzA1* (0.079 ± 0.01), *pvdA* (0.034 ± 0.04), *rhlA* (0.072 ± 0.017), and *pslB* (0.093 ± 0.01) in RRLP1 (Figure 6b). Similarly, the fold mRNA levels in PLD-treated RRLP2 were − 0.38 ± 0.04, 0.16 ± 0.05, 0.21 ± 0.07, 0.063 ± 0.02, 0.014 ± 0.01, and 0.36 ± 0.04 for *lasA*, *lasB*, *phzA1*, *pvdA*, *rhlA* and *pslB*, respectively (Figure 6c). These results were statistically significant in comparison with the respective growth control ($p < 0.001$).

Similarly, the genetic determinants that regulate the QS mechanism exhibited significant down-regulation in PLD-treated (3.5 mM) *P. aeruginosa* PAO1, RRLP1 and RRLP2 cells ($p < 0.001$) (Figure 7a–c). The expression of *lasI*, *lasR*, *rhlI*, *rhlR* and *mvfR* in PAO1 were down-regulated by 85.0%, 81.67%, 75.3%, 76.3% and 87.68%, respectively (Figure 7a). Additionally, tested clinical strains also exhibited significant % reduction in the relative mRNA levels of QS-associated genes *lasI* (65.67%), *lasR* (61.00%), *rhlI* (65.33%), *rhlR*

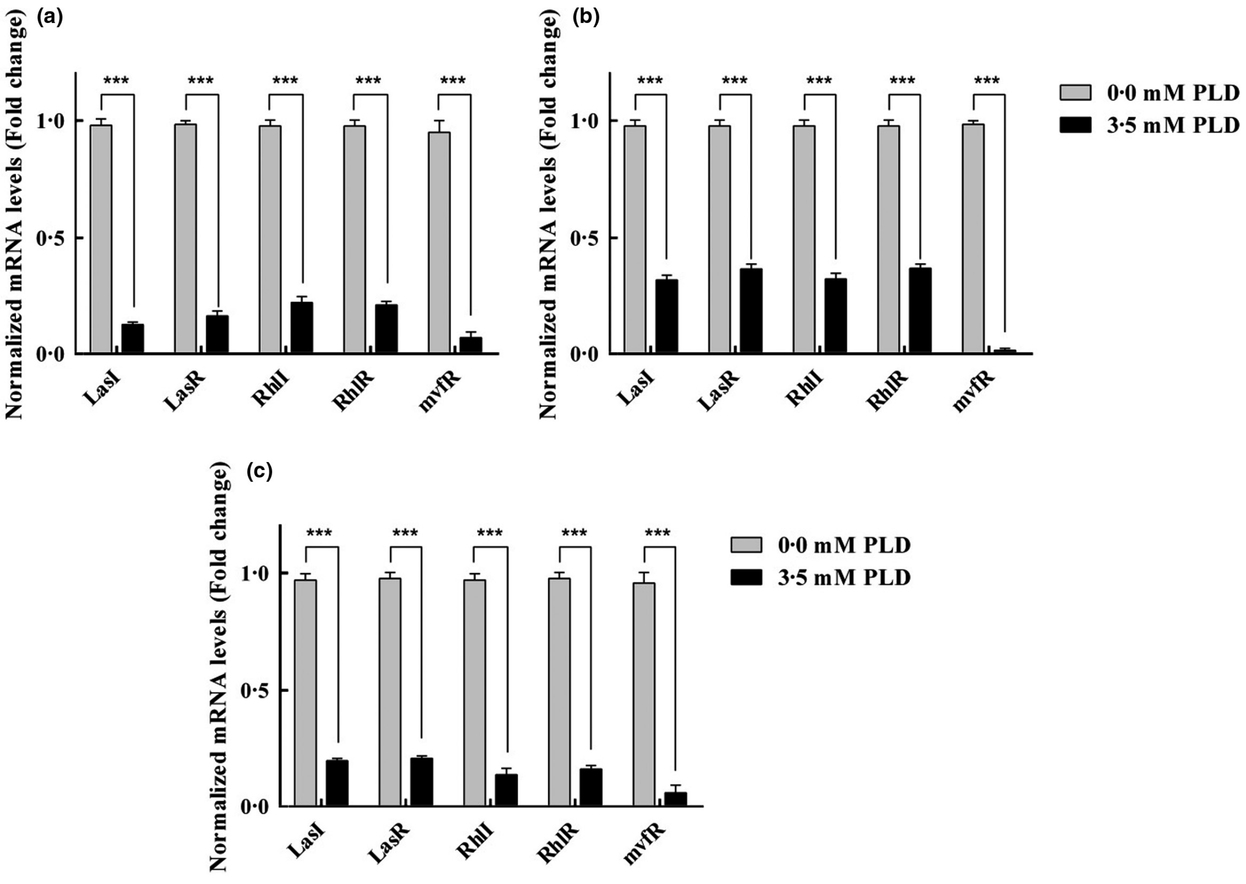

FIGURE 7 mRNA expression analysis of quorum sensing-associated genes after perillaldehyde (3.5 mM) treatment. (a) PAO1, (b) RRLP1 and (c) RRLP2. The results were represented as mean ± SD ($n = 4$) and analysed by two-way ANOVA with Sidak's multiple comparison post-test. Comparisons were performed as respective genes of perillaldehyde untreated (control) versus perillaldehyde treated (***$p < 0.001$). ANOVA, analysis of variance; SD, standard deviation

(60.67%) and *mvfR* (96.41%) in RRLP1 (Figure 7b). Similarly, a significant transcriptional level inhibition of 77.00% (*lasI*), 70.7% (*lasR*), 83.00% (*rhlI*), 81.3% (*rhlR*) and 89.4% (*mvfR*) was seen with PLD-treated RRLP2 cells (Figure 7c).

Molecular docking

The ligand PLD hypothetical binding pocket in PqsR (Figure 8). The sp2 carbonyl oxygen efficiently makes a hydrogen bonding interaction with Leu197. The cyclohexenyl and alkenyl part of the ligand is surrounded by strong hydrophobic residues like Ala102, Pro129, Ile149, Ala168, Leu207, Leu208, Ala237 and Pro238. The docking studies revealed that a combination of both hydrogen bonding and hydrophobic interaction could successfully dock the ligand into the active site of the inhibitor binding cavity.

Binding free energy analysis

The results obtained after calculation were approximate free energies of binding with a more negative value indicating a stronger binding. According to the MM-GBSA analysis, it was found that PLD had binding free energy of −68.44 kcal/mol, which was comparable to the co-crystallized ligand having −76.60 kcal/mol.

DISCUSSION

Biofilm is a compactly packed colony of microbial cells that with the help of secreted polymers and VFs attach and grow on living or non-living surfaces. The majority of the medical device surfaces are prone to biofilm formation and have given rise to elevated levels of the medical device associated with nosocomial infections. Biofilm formations on medical device surfaces perform a major part in the pathogenesis of biofilm linked infections as it disturbs host defence systems and delivers effortless contact of pathogens to the host. Therefore renewed interest has arisen in new therapeutic strategies for restricting biofilm establishment and other VFs (Das et al., 2016). The current study was focused on the effect of sub-MIC doses of PLD: a natural non-toxic food additive, on biofilm inhibition and anti-virulence property against the standard

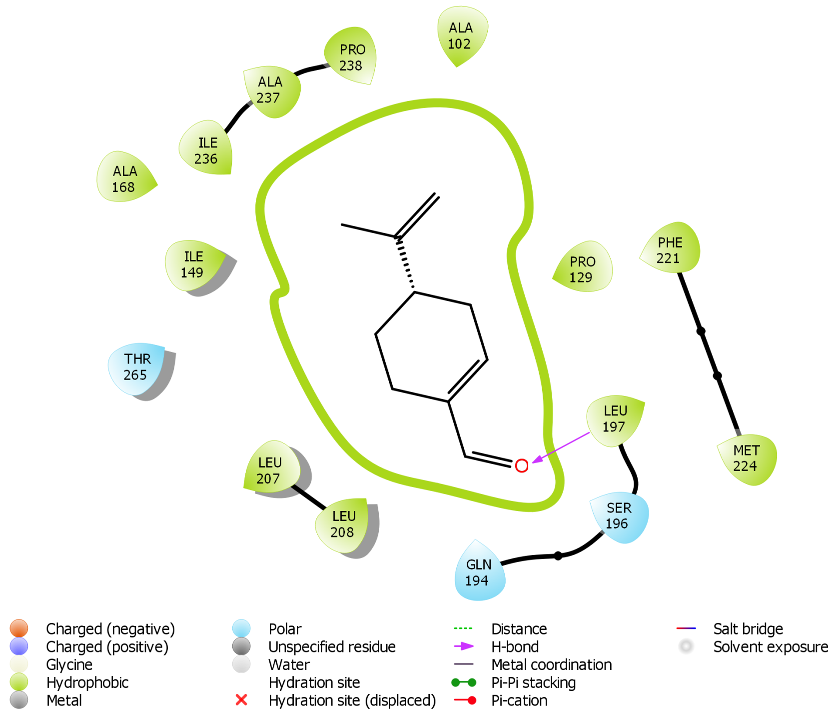

FIGURE 8 Binding mode of QS regulator protein (PqsR) from *Pseudomonas aeruginosa* with perillaldehyde (PLD). QS, quorum sensing

strain (*P. aeruginosa* PAO1) and two multidrug-resistant catheter isolates (*P. aeruginosa* RRLP1 and RRLP2).

Considering the worldwide circumstances of forever growing multi-drug resistant strains, biofilm inhibition and anti-virulence agents are desired to be formulated without any bactericidal or bacteriostatic effects against the tested organisms (Yan & Bassler, 2019). Accordingly, in the present study, the MIC of PLD was assessed against PAO1, RRLP1 and RRLP2 and all the further analyses were achieved at sub-MIC doses of PLD (2.5 and 3.5 mM). Dose–response growth curve results of PLD-treated (5.5 mM) test organisms confirmed an unaffected growth pattern (Figure S1) in comparison to the respective growth controls. Similar observations were reported by many researchers wherein the tested levels for biofilm inhibition and/or anti-virulence property were selected without any bactericidal or bacteriostatic effects (Abinaya & Gayathri, 2019; Chatterjee et al., 2017; Rajkumari et al., 2019; Rasamiravaka et al., 2017).

The microbial biofilm mode of growth provides a protective environment allowing evasion from the antimicrobial activity and host immune systems. Henceforth, biofilm inhibition is the primary objective for preventing biofilm-associated nosocomial infections like CAUTIs (Packiavathy et al., 2014; Rathinam et al., 2017). Anti-biofilm strategies may include prevention of biofilm establishment, upsetting the biofilm architecture and/or disorganizing preformed biofilms (Abinaya & Gayathri, 2019). The results of the biofilm biomass assay in the current study specified diminished biofilm biomass in all the PLD tested organisms (Figure 1). In addition, PLD resulted in a biofilm matrix with lesser complexity which resulted in a reduced number of associated cells, which was more seeming from the light microscopy images (Figure 2). This result was consistent with the earlier reports of plant derived compounds disrupting the bacterial biofilm formation without influencing planktonic growth (Abinaya & Gayathri, 2019; Chatterjee et al., 2017; Viswanathan, Rathinam, & Suneeva, 2015). Biofilm development begins with the reversible attachment of the planktonic bacteria to living or non-living surfaces. The progression of attachment is significantly affected by the CSH of the bacteria cells. Therefore, microbes typically settle on non-polar or hydrophobic surfaces (Rajkumari et al., 2019). The present study exhibited a significant lessening in the CSH of the sub-MIC doses of PLD tested organisms indicating its influence of initiating the biofilm formation, which might be the reason for the observed biofilm inhibition property. Likewise, She et al. (2018) also observed inhibition of CSH in the treatment with Meloxicam at the sub-lethal concentrations (She et al., 2018).

An additional mechanism of the anti-biofilm strategy is hampering the characteristic biofilm architecture (Ohman et al., 1980; Packiavathy et al., 2014; Rajkumari et al., 2019). In the present study, the scanning electron micrographs of PLD-treated biofilm on the urinary catheter pieces revealed that the architecture was looser and less thick in comparison with the PLD-untreated biofilms. In addition, PLD-treated biofilms were devoid of characteristic structural features which are in agreement with the earlier reports of Hnamte et al., (2019) with Mosloflavone and Abinaya and Gayathri (2019) with 3,5,7-trihydroxyflavone, in which a substantial reduction in biofilm arrangement, thickness and associated structural features were observed (Abinaya & Gayathri, 2019; Hnamte et al., 2019).

As a treatment strategy for biofilm-associated infections, anti-biofilm agent(s) should prevent further microbial adhesion to the pre-formed biofilms and reduce bacterial agglomeration. Recent studies on surfaces with poly(ethylene glycol) based multifunctional coating of the synthetic QS inhibitor exhibited reduced cell attachment and adhesion of pathogens like *S. aureus* and *P. aeruginosa* (Ozcelik et al., 2017). Similarly, urinary catheters coated with quorum quenching and matrix-degrading enzyme multilayers exhibited reduced bacterial fouling and subsequent biofilm formation by medically relevant pathogens (Ivanova et al., 2015). Corroborating with these reports, PLD exhibited biofilm inhibition and eradication of pre-formed biofilms. Biofilm eradication property of sub-MIC doses of PLD in preformed biofilms was evaluated through colony counting technique. In the present study, the tested dose of PLD (3.5 mM) was capable of reducing the count of biofilm-associated bacteria at 24 h ($p < 0.001$). In comparison to the PLD-untreated control biofilms, sub-MIC doses of PLD have significantly prevented further bacterial adhesion and accumulation onto the surface and prevented the maturation of biofilms, even after forming a conditioning film (Figure 3). Results of the present study exhibited the multiple mechanisms of PLD as inhibiting the biofilm formation, destabilizing biofilm architecture and disrupting preformed biofilms by the tested organisms.

Inhibition of extracellular polysaccharide production will facilitate a direct effect on biofilms as it helps in preserving the structural complexity of biofilm, and microcolony establishment. Furthermore, EPS deliberates antibiotic resistance to biofilm linked organisms by masking them from the antibiotics by acting as a protective barrier (Lotfy et al., 2018). Earlier studies have confirmed that reducing the EPS assembly will enable direct antibiotic exposure and induce rapid biofilm eradication (Abinaya & Gayathri, 2019; Rajkumari et al., 2019; Rathinam et al., 2017; She et al., 2018). In comparison to the PLD-untreated control biofilms, a sub-MIC dose of PLD (3.5 mM) has significantly ($p < 0.001$) prevented EPS production, aiding in hampering the biofilm architecture, which supported the anti-biofilm and SEM observations (Figure 5). The multifaceted phenotype of motility was associated with diverse regulatory constituents. Among the various VFs associated with biofilm formation in *P. aeruginosa*, swarming motility enact a major part which was regulated by Las and Rhl QS systems (Overhage et al., 2008). Subsequently, the disruption of motility due to certain modulation of rhamnolipids biosynthesis as well, in the case of *P. aeruginosa*. During biofilm growth, rhamnolipids are described to generate and uphold fluid channels for water and oxygen flow around the foundation of the biofilm. Additionally, they are significant for establishing three-dimensional structures in biofilms (Abinaya & Gayathri, 2019; Overhage et al., 2008). From Table 1, it is evident that the swarming motility of PAO1, RRLP1 and RRLP2 was significantly deprived of control when treated with sub-MIC doses of PLD. In addition, the present study also displayed the capability of PLD to significantly inhibit the production of biosurfactants (rhamnolipid) (Figure 5). Since the cellular mass translocation all through swarming migration depends on the expression of biosurfactant particles, whose expression is regulated by the QS mechanism, the obtained results suggested that PLD interfere with the QS circuit of the tested organisms. These results were in corroboration with previous reports of various phytocompounds exhibiting a significant reduction in motility and rhamnolipid production in *P. aeruginosa* PAO1 at their sub-MIC doses (Abinaya & Gayathri, 2019; Chatterjee et al., 2017; Das et al., 2016; Rajkumari et al., 2019; Rathinam et al., 2017; She et al., 2018). In general, biofilm-associated elements like psl polysaccharides contributes to the adherence, hydration, defence and inflexibility of the biofilm matrix, whereas, the amphipathic glycolipid, rhamnolipid eases microbial motility, upsurges the CSH and upholds open channels thus encouraging the micro-colonies formation (Rajkumari et al., 2019). Therefore, it may be confirmed that PLD has driven the down-regulation of polysaccharide and rhamnolipids biosynthesis, has distraught the rigidity of matrix and cellular fitness, consequently hampering the structural as well as the functional stability of PAO1 biofilms.

For effective pathogenesis, besides biofilm formation, extracellular VFs secretion also plays an important role by enhancing adhesion, invasion and colonization of the host tissue (Fleitas Martínez et al., 2019). In *P. aeruginosa* production of these VFs are under the regulation of the QS mechanism and they interfere with multicellular functions in the host for successfully establishing the pathogen (Aswathanarayan & Ravishankar Rai, 2015). Quantification of these quorum regulated VFs not only affirms the anti-virulence properties of the agent but also suggests that the mechanism is via QS inhibition. In the present study, it was noticed that sub-MIC doses of PLD exhibited a pronounce interference in the production of QS regulated VFs like pyocyanin, pyoverdine, lasA protease and lasB elastase (Figure 5a–c). Proteases and elastases are hydrolytic enzymes that slice a diversity of host proteins and are controlled by Las system. Similarly, pyocyanin and pyoverdine are other major

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VFs required to sense and sequester iron from the environment (Fleitas Martínez et al., 2019; Viswanathan, Suneeva, & Rathinam, 2015). The observed reduction of these VFs suggests ineffectual pathogenesis without generating any selection pressure. In addition, the reduction in VFs in concordance with the anti-QS effect of sub-MIC doses of PLD by interfering with the QS-associated genes that are involved in regulating the production of these VFs.

With respect to the observed results with the reduction in levels of VFs, the effect of PLD (3.5 mM) incorporation on the expression of the respective virulence determinants like *lasA* (encoding protease), *lasB* (encoding elastase), *phzA1* (encoding pyocyanin), *pvdA* (encoding pyoverdine), *rhlA* (encoding rhamnosyl transferase), *pslB* (encoding psl biofilm polysaccharide) for PAO1, RRLP1 and RRLP2 were analysed (Figure 6a–c). The RT-qPCR data of the present study demonstrated that PLD has significantly ($p < 0.001$) down-regulated the relative expression of all tested VFs determinants. Down-expression of these genes may interfere with the normal production of VFs as observed in the present study, which corroborated well with the previous findings (Abinaya & Gayathri, 2019; Chatterjee et al., 2017; Hnamte et al., 2019; Ivanova et al., 2015; Ozcelik et al., 2017; Rasamiravaka et al., 2015; Rasamiravaka et al., 2017; Rathinam et al., 2017; She et al., 2018).

The previous study from our laboratory has identified the competitive binding of phytocompound; eugenol to QS receptor(s) as the responsible mechanism for its anti-virulence property at sub-MIC doses (Rathinam et al., 2017). Further, in the present study, 3.5 mM of PLD considerably repressed the expression of the QS receptors (*lasR*, *rhlR* and *mvfR*) as well as autoinducer synthase (*lasI* and *rhlI*) in PAO1, RRLP1 and RRLP2 cells (Figure 7a–c). In the recent findings, a significant arrest in the expression of *lasR*, *rhlR* and *mvfR* was attained in the presence of Meloxicam (Das et al., 2016). These obtained results were also similar to the report wherein the expression of *lasI*, *lasR*, *rhlI* and *rhlR* were down-regulated in the presence of 0.300 mM of equisetin (Hodgkinson et al., 2012). Based on these results, it was confirmed that with PLD (3.5 mM) the possible mechanism of the observed biofilm inhibition and anti-virulence property is through QS inhibition (QSI) since, in these tested organisms the VF genes (*lasA*, *lasB*, *phzA1*, *pvdA*, *rhlA* and *pslB*) are regulated by the QS system.

The majority of the natural and synthetic agents attain the QS inhibition through the deactivation of a signalling cascade by binding to the active site of respective receptors. This notion is supported by various studies on synthetic and natural signal mimics which are found to inhibit the QS signal cascade by binding to the respective ligand-binding site (Hodgkinson et al., 2012; McInnis & Blackwell, 2011; Viswanathan, Rathinam, & Suneeva, 2015). Further, in the present study, we examined the molecular interactions of QS-regulating protein PqsR with PLD (Figure 8). Amino acids present in the binding sites of the PqsR protein play crucial roles in the dimerization and activation of the receptor. Based on our results, we propose that the binding of PLD to the amino acids might be directly responsible for the inactivation of the PqsR protein, resulting in subsequent repression of QS and QS-regulated VFs. We speculate that this occurs either by preventing protein dimerization or there are direct effects on the DNA resulting in suppression of gene expression downstream in the QS system.

Comprehensive structure–activity relationship studies of QS receptors vs. autoinducer mimics concluded that the homoserine lactone moiety (present in Autoinducer) substituted with an aromatic system is found to have augmented the ligand's antagonistic properties (Hodgkinson et al., 2012; McInnis & Blackwell, 2011). Interestingly, PLD being a monoterpenoid compound containing an aldehyde functional group shared significant structural similarity with previously reported terpenoids (oleanolic aldehyde coumarate, ursolic acid, oleanolic acid, cassipourol and β-sitosterol) from various essential oils with potent anti-QS properties (Kiplimo et al., 2011; Rajkumari et al., 2019; Rasamiravaka et al., 2017). The obtained results are concordant with the previous results by Abinaya & Gayathri, 2019; Hnamte et al., 2019; Rajkumari et al., 2019.

The significance of the study is in finding the biofilm inhibition, biofilm eradication and anti-virulence ability of PLD: a natural non-toxic food additive from Perilla. This study further evaluated the QSI property of sub-MIC doses of PLD not only against *P. aeruginosa* PAO1 (reference strain) but also in catheter isolates (*P. aeruginosa* RRLP1 and RRLP2) from patients with CAUTIs. Furthermore, this study provided insights into the molecular mechanism of the anti-QS property of PLD via interacting with the QS-regulating protein, PqsR. Perillaldehyde, being a major active principle in many essential oils, is considered safe for human use as a natural food additive. In the light of the in vitro results from the present study, PLD can be a suitable candidate for countering biofilms and associated pathogens. Necessary in vivo studies are needed to confirm its contribution towards the prevention of biofilm-associated nosocomial infections like CAUTIs.

ACKNOWLEDGEMENTS

E.K.R. thanks VIT University, Vellore for providing the research facility and VIT SEED GRANT (RGEMS) for providing the financial support. P.R. is thankful to the management, Pushpagiri Medical sciences and research centre, Thiruvalla, for permitting to use of the facilities at Pushpagiri Research Centre and the financial support for performing the study.

CONFLICT OF INTEREST

None declared.

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SUPPORTING INFORMATION

Additional supporting information may be found in the online version of the article at the publisher's website.

How to cite this article: Benny, A.T., Rathinam, P., Dev, S., Mathew, B. & Radhakrishnan, E.K. (2022) Perillaldehyde mitigates virulence factors and biofilm formation of *Pseudomonas aeruginosa* clinical isolates, by acting on the quorum sensing mechanism in vitro. *Journal of Applied Microbiology*, 133, 385–399. Available from: https://doi.org/10.1111/jam.15565